

Programme: B.E – CSE (AI&ML) and CSE(Cyber Security)
Internal Assessment – I

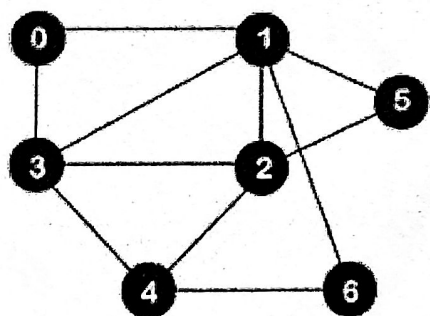
TERM: EVEN SEM 2023-2024	COURSE NAME: Design and Analysis of Algorithms
DATE: 31/05/2024 TIME: 9:30am to 10:30am	COURSE CODE: CI43/CY43
MAX MARKS: 30	PORTIONS: L1-L20
SEMESTER: 4 th SEM	SECTION: A



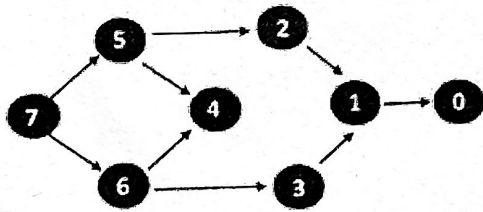
Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels	CO	Marks
1.a	Consider the scenario given. There are n people numbered from 1 to n in a circle. Starting from person 1, we eliminate every second person until only survivor is left. Design an efficient algorithm to find the survivor's number $J(n)$. Calculate the time complexity class for the algorithm you defined.	L4	CO1	4
b	Consider the following recursive algorithm. ALGORITHM $Q(n)$ //Input: A positive integer n if $n = 1$ return 1 else return $Q(n - 1) + 2 * n - 1$ i. Set up a recurrence relation for the algorithm and solve it to determine what this algorithm computes. ii. Prove the time complexity using Masters Theorem.	L4	CO2	5
c	Consider the graph given in Fig 1.c.  Fig 1.c Undirected Graph Starting at vertex 0 and resolving ties by the vertex alphabetical order, obtain the following. i. Traverse the graph by depth-first search. ii. Construct the corresponding depth-first search tree. iii. Give the order in which the vertices were reached for the first time (pushed onto the traversal stack) and the order in which the vertices became dead ends (popped off the stack).	L3	CO3	6
2. a	Consider the given scenario: Friends Dal and Sean want to take a car trip across the country from Bangalore to Delhi by driving between cities during the day, and staying at a hotel in some city each night. There are n cities across the country. For each city c_i , Dal and Sean have compiled:	L4	CO3	4



	• a list L_i of the at most 10 other cities they could drive to in a single day starting from city c_i. Illustrate all the possible routes to determine whether it is possible for Dal and Sean to drive from Bangalore to Delhi, by defining a graph traversal algorithm, considering visiting one city only once.			
b	Solve the following i. Consider the following $f(n)$ and $g(n)$ $f(n) = 3n + 2$ $g(n) = n$ Prove that $f(n) \in O(n)$. Compute c and n_0. ii. Consider the following $f(n)$ and $g(n)$ $f(n) = 3n^2 + 24n + 148$ $g(n) = n^2$ Prove that $f(n) \in O(n^2)$. Compute c and n_0. iii. Compute the time complexity of the code snippet. <pre>for (int i = 0; i < n; i++) for (int j = i; j > 0; j--) cout << i << j;</pre> iv. Compute the time complexity of the code snippet. <pre>int calcFactorial(n) { int factorial = 1; for (int i = 2; i <= n; i++) { factorial = factorial * i; } return factorial; };</pre>	L3	CO1	2+2+1 +1=6
c	Apply DFS-based topological sort algorithm and obtain the topological sort order for the graph given in Fig.2c. Show the traces of stack operations.  Fig 2c. Digraph	L3	CO3	5
3.	Construct a heap for the list 1, 8, 6, 5, 3, 7, 4 by the bottom-up algorithm. Sort the following lists by heapsort by using the array representation of heaps.	L3	CO2	7
a	Is heapsort a stable sorting algorithm?			
b	Solve the following recurrence relations using substitution method. a. $x(n) = x(n-1) + 5$ for $n > 1$, $x(1) = 0$ b. $x(n) = x(n/3) + 1$ for $n > 1$, $x(1) = 1$	L3	CO1	4
c	A bank confiscates n bank cards suspecting them to be involved in fraud. Each card corresponds to a unique account but an account can have many cards corresponding to it. Therefore, two bank cards are equivalent if they belong to the same account. The bank has a testing machine for finding out if two cards are equivalent. They want to know that among the collection of n cards, are there more than $n/2$ cards that are equivalent. The only operation that the machine can do is to select two cards and tests them for equivalence. You are required to come up with a solution to this using only $O(n \log n)$ invocations of the testing machine.	L4	CO2	4